Assignment-03

D D M M Y Y Y Y

01) A die was thrown 9000 times and throw of 5 or 6 was obtained 3240 times on the assumption of random throwing do the data indicate an unbiased die?

→ 1, 2, 3, 4, (5), (6)

probability of getting 5 (or) 6 in a single throw is

$p = 2/6$ $p + q = 1$

$p = 1/3$ $q = 2/3$

Expected no. of success

$np = 9000 \times 1/3$

$np = 3000$

Observed no. of success (3 or 4)

$x - np = 3240 - 3000$

$x - np = 240$

$$z = \frac{x - np}{\sqrt{npq}}$$

$$= \frac{240}{\sqrt{2000}}$$

$z = 5.366 > 2.58$

As $z > 2.58$ the hypothesis is rejected at 1% level of significance. Hence we conclude that the die is unbiased.

02) Before an increase in excise duty on tea 400 people out of a sample 500 persons were found to be a tea drinkers. After an increase in duty 400 people were tea drinkers in a sample of 600 people. Using standard error of proportion, state whether there is a significant decrease in the consumption of tea for 95% and 99% level of significance?

→ Null hypothesis; $H_0: P_1 = P_2$

Alternative hypothesis; $H_1: P_1 \neq P_2$

$n_1 = 500$ $n_2 = 600$

$x_1 = 400$ $x_2 = 400$

In 500 people out of 400 people found to be headrenkers

$$P_1 = \frac{400}{500} = 0.8$$

In 600 people out of 400 people found to be headrenkers

$$P_2 = \frac{400}{600} = 0.6666$$

$$P = \frac{x_1 + x_2}{n_1 + n_2}$$

$$= \frac{400+400}{500+600}$$

$$= \frac{800}{1100}$$

$$P = 0.7272$$

$$P + Q = 1$$

$$Q = 1 - 0.7272$$

$$Q = 0.2728$$

$$e = \sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

$$= \sqrt{0.7272\,(0.2728)\left(\frac{1}{500} + \frac{1}{600}\right)}$$

$$= \sqrt{0.19838016\left(\frac{11}{3000}\right)}$$

$$= \sqrt{0.000727339392}$$

$$e = 0.02697$$

$$Z = \frac{P_1 - P_2}{e} = \frac{0.8 - 0.6666}{0.02697}$$

$$Z = 4.9462 > 1.96 \text{ \& } > 2.58$$

AS Z is $Z > 1.96$ & $Z > 2.58$ the hypothesis is rejected at

1% & 5% level of significance.
Hence we conclude that there is a significant difference b/w two populations.

03) A survey is conducted in a slum locality of 2000 families by selecting a sample of size of 800 families. It was revealed that 180 families were illiterate. Find the probable limits of the illiterate families in the population of 2000.

→ Probable of illiterate families $p = 180/800$

$p = 0.225$

$p + q = 1$

$q = 0.775$

Sample size $n = 800$

probability limits $= p \pm 2.58\sqrt{\frac{pq}{n}}$

$= 0.225 \pm 2.58\sqrt{\frac{0.225(0.775)}{800}}$

$= 0.225 \pm 0.0380$

$0.225 + 0.0380$ & $0.225 - 0.0380$

0.187 & 0.263

∴ probable limit of illiterate family in the population of 2000 is $= 0.187 \times 2000$ & 0.263×2000

$= 374$ & 526

Thus 374 & 526 families probable limits.

04) In 324 throws a die an odd number turned up 181 times. It is reasonable to think that the dice is unbiased.

→ The no. of throws $n = 324$

The probability of die of 6 phased dice turned into odd no.

$p = 1/2 \quad q = 1/2$

Observed no. of turned throws $x = 181$

Expected no. of throws $np = 324 \times \frac{1}{2}$ $np = 162$

difference $x - np = 181 - 162 = 19$

$$z = \frac{x - np}{\sqrt{npq}}$$

$$= \frac{19}{\sqrt{162 \times \frac{1}{2}}}$$

$$z = 2.111 > 1.96$$

As $z > 1.96$ the hypothesis is rejected at 5% level of significance

Hence the die is unbiased.

05) The mean weight obtained from a random sample of size 100 is 64gms. The standard deviation of the weight distribution of the population is 3gms. Test the statement that the mean weight of population is 67gms. at 5% level of significance. Also set up 99% confidence limits of the mean weight of the population.

→ sample size, $n = 100$

$$\mu = \bar{x} = 64 \text{ gms}$$

$$\sigma = 3 \text{ gms}$$

$$\mu_0 = 67 \text{ gms}$$

5% level $\alpha = 0.05$

Null Hypothesis, $H_0 : \mu_1 = \mu_2$

Alternative Hypothesis, $H_1 : \mu_1 \neq \mu_2$

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

$$= \frac{64 - 67}{3 / \sqrt{100}}$$

$$z = -10$$

D D M M Y Y Y Y

$z = -10$

As $z > -1.96$ the hypothesis is ~~accepted~~ rejected at 5% level of significance.

For 99% $\Rightarrow z = 2.58$

Confidence limits $= \mu \pm z\left(\frac{\sigma}{\sqrt{n}}\right)$

$C.L = 64 \pm 2.58\left(\frac{3}{\sqrt{100}}\right)$

$= 64 \pm 0.774$

$= (64 + 0.774, 64 - 0.774)$

$= (64.774, 63.226)$

Confidence limits $= (64.774, 63.226)$

D D M M Y Y Y Y

06) In a sample of 100 people in this city, the average income was Rs 210, with a standard deviation of Rs. 10. For another sample of 150 persons, the average income was Rs. 220, with a standard deviation of Rs. 12. The S.D of the incomes of the people of the city was Rs. 11. Test whether there is any significant d/f b/w the average incomes of the probabilities.

→ H_0 : $\mu_1 = \mu_2$ Null Hypothesis

H_1 : $\mu_1 \neq \mu_2$ Alternative Hypothesis

$\bar{x}_1 = 210$ $\bar{x}_2 = 220$ $\sigma = 11$

$n_1 = 100$ $n_2 = 150$

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}}}$$

$$= \frac{210 - 220}{\sqrt{\frac{121}{100} + \frac{121}{150}}}$$

$$= -7.041$$

$z = -7.041$

As $z > -1.96$ the hypothesis is rejected at 5% level of significance.

∴ Hence there is a significant diff? between the average incomes of the probabilities.

D D M M Y Y Y Y

07) Define Null hypothesis, significance level, critical region, Type-I and Type-II errors in a statistical test.

→ Null hypothesis:

A statistical hypothesis which we formulate to check whether to accept or rejected is called null hypothesis.

Significance level

Probability of (Type I error) is called level of significance.

Critical region

The region in which the calculated sample value falling is rejected is called critical region.

Type-I error

Rejecting H_0 when it is true is called Type-I error.

Type-II error

Accepting H_0 when it is false is called Type-II error.

08) In a sample of 600 men from a certain city, 450 are found smokers. In other sample of 900 men from another city, 450 are smokers. Do the indicate that the cities are significantly different with respect to the habit of smoking among men. Test at 5% significance level.

→ Null hypothesis: $H_0 : P_1 = P_2$

Alternative hypothesis: $H_1 : P_1 \neq P_2$

$n_1 = 600$ $n_2 = 900$

$x_1 = 450$ $x_2 = 450$

In 600 people 450 are found to be smokers

$$P_1 = \frac{450}{600} = 0.75$$

In 900 people 450 are found to be smokers

$$P_2 = \frac{450}{900} = 0.5$$

D D M M Y Y Y Y

$$P = \frac{x_1 + x_2}{n_1 + n_2}$$

$$= \frac{450 + 450}{600 + 900}$$

$$P = 0.6$$

$$P + Q = 1$$

$$Q = 0.4$$

$$e = \sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

$$= \sqrt{0.6(0.4)\left(\frac{1}{600} + \frac{1}{900}\right)}$$

$$e = 0.02581$$

$$z = \frac{p_1 - p_2}{e}$$

$$= \frac{0.75 - 0.5}{0.02581}$$

$$z = 9.6861 > 1.96$$

As $z > 1.96$ the hypothesis is rejected at 5% level of significance.

09) ~~Explain~~ A coin was tossed 4000 times and head turned up 216 times. Test the hypothesis that the coin is unbiased at 5% level of significance.

→ The coin is unbiased

$p = 1/2 \quad q = 1/2$

Observed no. of head turned up $x = 216$

no. of coin tossed times $n = 4000$

Expected no. of head turned times $np = 4000 \times \frac{1}{2} = 2000$

difference $x - np = 216 - 2000 = 16$

$$z = \frac{x - np}{\sqrt{npq}}$$

$$Z = \frac{16}{\sqrt{400 \times \frac{1}{2} \times \frac{1}{2}}}$$

$$= \frac{16}{\sqrt{100}}$$

$$= \frac{16}{10}$$

$$Z = 1.6 < 1.96$$

$Z < 1.96$ the hypotheses is accepted at 5% level of segnefecance.

(10) In a cety A 20% of a random sample of 900 school boys had a slight physecal defect. In another cety B, 18.5% of a random sample of 1600 school boys had the same defect. Is the deffernce b/w the proportions segnefecant at 5% segnfecant level.

→ Null hypotheses $H_0 : P_1 = P_2$

Alternate hypotheses $H_1 : P_1 \neq P_2$

$n_1 = 900$ $n_2 = 1600$

$x_1 = 20\%$ $x_2 = 18.5\%$

900, 20% $x_1 = \frac{20}{100} \times 900$ $x_1 = 180$

1600, 18.5% $x_2 = \frac{18.5}{100} \times 1600$ $x_2 = 1600$ 296

$$P = \frac{x_1 + x_2}{n_1 + n_2}$$

$$= \frac{180 + 296}{900 + 1600}$$

$$P = 0.1904$$

$$P + Q = 1$$

$$Q = 0.8096$$

DD MM YYYY

Assignment - 04

01) An experiment on Pea breeding the following frequency of seeds were obtained

Round & Yellow	Wrinkled & Yellow	Round & Green	Wrinkled & Green	Total
315	101	108	32	556.

Theory predicts that the frequences should be in the proportions 9:3:3:1. Examine the correspondence b/w theory & experiment ($\chi^2_{0.05} = 7.815$).

→ Total no. of seeds = 556

Sum of the ratio = 9+3+3+1 = 16

$$E(A) = 556 \times \frac{9}{16} = 312.75 = 313$$

$$E(B) = 556 \times \frac{3}{16} = 104.25 = 104$$

$$E(C) = 556 \times \frac{3}{16} = 104$$

$$E(D) = 556 \times \frac{1}{16} = 34.75 = 35$$

O_i	315	101	108	32
E_i	313	104	104	35
$(O_i - E_i)^2$	4	9	16	9

$$\chi^2 \sum_{i=1}^{n} = \frac{(O_i - E_i)^2}{E_i}$$

$$= \frac{4}{313} + \frac{9}{104} + \frac{16}{104} + \frac{9}{35}$$

$\chi^2_{cal} = 0.51$

$\chi^2_{tab} = 7.815$

$\chi^2_{cal} < \chi^2_{tab}$

D D M M Y Y Y Y

02) Use the central Limit theorm to evaluate $P[50 < \bar{x} < 56]$ where $\bar{x}$ represents the mean of a random sample of size 100 from an infinite population with mean $\mu = 53$ and variance $\sigma^2 = 400$ (Given, $A(1.5) = 0.4332$).

→ $P[50 < \bar{x} < 56]$ $\bar{x} \Rightarrow \sigma^2 = 400$ $\sigma = 20$ $n = 100$ $\mu = 53$

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$= \frac{\bar{x} - 53}{20/\sqrt{100}}$$

$$z = \frac{\bar{x} - 53}{2}$$

At $\bar{x} = 50$ $\bar{x} = 56$

$$z = \frac{50 - 53}{2} \qquad z = \frac{56 - 53}{2}$$

$z = -1.5$ $z = 1.5$

$$P[50 < \bar{x} < 56] = P[1.5 < \bar{x} < -1.6]$$

$$= 2P[0 \leq z \leq 1.5]$$

$$= 2P(1.5)$$

$$= 2 \cdot \phi(1.5)$$

$$= 2 \times 0.4332$$

$$P[50 < \bar{x} < 56] = 0.8664$$

D D M M Y Y Y Y

03) Ten individuals are chosen at random from a population and their heights in inches found to be 63, 63, 66, 67, 68, 69, 70, 70, 71 and 71. Test the hypotheses that the mean height of the universe is 66 inches. ($t_{0.05} = 2.262$ for 9 df)

→ $n = 10$ μ (population mean) $= 66$

$$\bar{x} = \frac{63+63+66+67+68+69+70+70+71+71}{10}$$

$$\bar{x} = 67.8$$

Ho: there is no diff^{n} in sample & universe Mean 66 inches.

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

$$= \frac{1}{9} [81.6]$$

$$S^2 = 9.0667$$

$$S = 3.01$$

$$t = \frac{\bar{x} - \mu}{S} \sqrt{n}$$

$$= \frac{(67.8 - 66)}{3.01} \sqrt{10}$$

$$t = 1.8902$$

$$t_{cal} = 1.8902$$

$$t_{tab} = 2.262$$

$$t_{cal} < t_{tab}$$

The hypothesis is accepted.

D D M M Y Y Y Y

04) The following table shows the runs scored by two batsman can it be said that the performance of batsman A is more consistent than the performance of batsman B? Use 1% level of significance ($F_{0.01, 4, 7} = 7.85$).

Batsman - A	40	50	35	25	60	70	65	55
Batsman - B	60	70	40	30	50			

→ H_0 : performance of batsman 1 is equal to batsman 2.

H_1 : Performance of batsman 1 is more consest than batsman 2

x_1	$(x_1 - \bar{x}_1)^2$	x_2	$(x_2 - \bar{x}_2)^2$	
40	100	60	100	
50	0	70	400	$n_1 = 8$
35	225	40	100	$n_2 = 5$
25	625	30	400	
60	100	50	0	
70	400	$\Sigma \bar{x}_2 = 250$	$\Sigma (x_2 - \bar{x}_2)^2 = 1000$	
65	225			
55	25			

$\Sigma \bar{x}_1 = 400$ $\Sigma (x_1 - \bar{x}_1)^2 = 1700$

$$\bar{x}_1 = \frac{400}{8} \left(\frac{\Sigma x_1}{n} \right) = 50$$

$$\bar{x}_2 = \frac{\Sigma x_2}{n_2} = \frac{250}{50} = 50$$

$$S_1^2 = \frac{\Sigma (x_1 - \bar{x}_1)^2}{n_1 - 1} = \frac{1700}{7} = 242.85$$

$$S_2^2 = \frac{\Sigma (x_2 - \bar{x}_2)^2}{n_2 - 1} = \frac{1000}{6} = 250$$

$$F = \frac{S_1^2}{S_2^2} = \frac{250}{242.85} = 1.0294 < 7.85$$

$F_{cal} < F_{tab}$ the hypothesis is accepted. The performance of bots batsmans is equal

05) The following table gives the no. of aircraft accidents that occured during the various days of the week. Find whether the accidents are uniformly distributed over the week?

Days	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Total
Accidents	14	16	8	12	11	9	14	84

Given $\chi^2_{0.05} = 12.59$.

→ Ho: The accidents are uniformly distributed over the week.

H1: The accidents are not uniformly distributed over the week

Estimated no. of accidents

E (Monday) = N/n = 84/7 = 12

E (Tue) = 12

E (Wed) = 12

E (Thu) = 12

E (Fri) = 12

E (Sat) = 12

E (Sun) = 12

Day	Sun	Mon	Tue	Wed	Thu	Fri	Sat
O_i	14	16	8	12	11	9	14
E_i	12	12	12	12	12	12	12
$(O_i - E_i)^2$	4	16	16	0	1	9	4

$$\chi^2 = \frac{\sum (O_i - E_i)^2}{E_i} = \frac{4}{12} + \frac{16}{12} + \frac{16}{12} + 0 + \frac{1}{12} + \frac{9}{12} + \frac{4}{12}$$

$\chi^2_{cal} = 4.1667$

$\chi^2_{tab} = 12.59$

$\chi^2_{cal} < \chi^2_{tab}$

∴ The accidents are uniformly distributed over the week.

D D M M Y Y Y Y

06) Consider the sample consisting of nine numbers 45, 47, 50, 52, 48, 47, 49, 53, 51. The is drawn from a population whose mean is 47.5. Find whether the sample mean differs significantly from the population mean at 5% level of significance ($t_{0.05}$ for 8 d.f = 2.31).

$\bar{x} = 49.111$ μ(population) = 47.5 $n = 9$

$H_0 : \mu = 47.5$ Null hypothesis

$H_1 : \mu \neq 47.5$ Alternative hypothesis

$$s^2 = \frac{1}{(n-1)} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

$$= \frac{1}{8} \{ 54.889$$

$s^2 = 6.86125$

$s = 2.6193$

$$t = \frac{\bar{x} - \mu}{s} \sqrt{n}$$

$$= \frac{49.111 - 47.5}{2.6193}$$

$t_{cal} = 0.615D$

$t_{tab} = 2.31$

$t_{cal} < t_{tab}$

∴ The hypothesis is accepted.

∴ The sample mean does not differ from population mean.

07) A random sample of size 25 from normal distribution $N(\mu, \sigma^2 = 4)$ yields, sample mean $\bar{x} = 78.3$. Obtain a 99% confidence interval for μ.

→ $n = 25 \quad \bar{Y} = 78.3 \quad \sigma^2 = 4 \quad \sigma = 2$

CI for 99% significance level $Z = 2.58$

$$CI = \mu \pm Z\left(\frac{\sigma}{\sqrt{n}}\right)$$

$$= 78.3 \pm 2.58\left(\frac{2}{\sqrt{25}}\right)$$

$$= 78.3 \pm 1.032$$

$$= [78.3 + 1.032, \; 78.3 - 1.032]$$

$$CI = (77.268, 79.332)$$

08) A survey of 320 families with 5 children each revealed the following distribution.

NO. of boys	5	4	3	2	1	0
NO. of girls	0	1	2	3	4	5
NO. of families	14	56	110	88	40	12

Is the result consistent with the hypothesis that Male and female births are equally probable at 5% level of significance

→ $N = 320 \quad K = 6 \quad p = 0.5, \; q = 0.5 \quad n = 5$

H_0: The probability of female & Male birth is equal.

H_1: The probability of female & Male birth is not equal.

$$Eq = N\left({}^{n}C_x \, p^x \, q^{n-x}\right)$$

$$Eq = N\left({}^{n}C_x \, p^x q^{n-x}\right)$$

$$= 320\left({}^{5}C_x \, (0.5)^x \, (0.5)^{5-x}\right)$$

$$Eq = 320\left[{}^{5}C_x \, (0.5)^5\right)$$

$$Eq = 320 \; {}^{5}C_x \, (0.03125) \quad Eq = {}^{5}C_x \cdot 10$$

DD MM YYYY

$E_i = 10 \cdot S_{Cx}$

$E_0 = 10 \cdot S_{C0} = 10 \times 1 = 10$

$E_1 = 10 \cdot S_{C1} = 10 \times 5 = 50$

$E_2 = 10 \cdot S_{C2} = 10 \times 10 = 100$

$E_3 = 10 \cdot S_{C3} = 10 \times 10 = 100$

$E_4 = 10 \cdot S_{C4} = 10 \times 5 = 50$

$E_5 = 10 \cdot S_{C5} = 10 \times 1 = 10$

No. of ~~boys~~ girls	0	1	2	3	4	5
No. of ~~girls~~ boys	5	4	3	2	1	0
O_i	14	56	110	88	40	12
E_i	10	50	100	100	50	10
$(O_i - E_i)^2$	16	36	100	144	100	4

$$\chi^2 = \frac{\Sigma (O_i - E_i)^2}{E_i}$$

$$= \frac{16}{10} + \frac{36}{50} + \frac{100}{100} + \frac{144}{100} + \frac{100}{50} + \frac{4}{10}$$

$\chi^2_{cal} = 7.16$

observed no. $n = 6$

$d.f = n - 1 = 5$

For 5% level of significance 0.05 over d.f(5)

$(5, 0.05)$

$\chi^2_{tab} = 11.07$

$\chi^2_{cal} < \chi^2_{tab}$

The hypotheses is accepted.

DD MM YYYY

08) A random sample of size 64 is taken from an infinite population having mean 112 and variance 144. Using central limit theorm, find the probability of getting the sample mean $\bar{x}$ greater than 114.5.

→ $n = 64 \quad \mu = 112 \quad \sigma^2 = 144 \quad \sigma = 12$

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$= \frac{\bar{x} - 112}{12/\sqrt{64}}$$

$$z = \frac{\bar{x} - 112}{1.5}$$

$\bar{x} = 114.5$

$$z = \frac{114.5 - 112}{1.5}$$

$z = 0.0223$

$z = 1.667$

$P(\bar{x} > 114.5) = P(z > 1.667)$

$= 0.5 - P(0 < z < 1.667)$

$= 0.5 - \phi(1.667)$

$= 0.5 - 0.4514$

$P(\bar{x} > 114.5) = 0.0489$

09) Let the observed value of the mean $\bar{x}$ of a random sample of size 20 from a normal distribution with mean μ and variance $\sigma^2 = 80$ be 81.2. Find a 90% and a 95% confidence intervals for μ.

→ $n = 20 \quad \sigma^2 = 80 \quad \mu = 81.2 \quad \sigma = 8.944$

$CI = z \pm \mu(\sigma/\sqrt{n})$ (for 90%)

$= 1.645 \pm 81.2(8.944/\sqrt{20})$

$= 1.645 \pm 162.395$

CI = [1.645 + 162.395, 1645 − 162.395]

= (164.04, −160.75)

For 95% Z = 1.96

CI = $\bar{x} \pm Z(\sigma/\sqrt{n})$

= ~~1.96~~

= 1.96 ± 81.2 (8.944/√20)

= 1.96 ± 162.395

CI = (164.355, −160.435)

10) An unknown destribution has a mean of 90 and S.D of 15. Samples of size n = 25 are drawn randomly from the population. Find the probability that the sample mean is b/w 85 & 92.

→ n = 25 μ = 90 σ = 15 P(85 ≤ 92)

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$= \frac{\bar{x} - 90}{15/\sqrt{25}}$$

$$Z = \frac{\bar{x} - 90}{3}$$

$\bar{x} = 85$ $\bar{x} = 92$

$Z = \frac{85-90}{3}$ $Z = \frac{92-90}{3}$

Z = −1.667 Z = 0.667

P(85 < $\bar{x}$ < 92) = P(−1.667 < $\bar{x}$ < 0.667)

= P(−1.667 < Z < 0) + P(0.667 < Z < 0)

= ϕ(+1.667) + ϕ(0.667)

= 0.4515 + 0.2454

P(85 < $\bar{x}$ < 92) = 0.6969

DD MM YYYY

Assignment -05

01) A manufacturing company has purchase three new machines of different brands and wishes to determine whether one of them is faster than the others in producing a certain output 3 hourly production figures are obtained at random from each other machine and the results are given below:

observation	A	B	C
1	25	31	24
2	30	39	30
3	36	38	28
4	38	42	25
5	31	35	28

Use ANOVA and determine whether the machine are significantly different in their mean speed. Given at 5% level $F_{2,12} = 3.89$

A	B	C	
25	31	24	
30	39	30	$k = 3$
36	38	28	$T = 480$
38	42	25	$N = 15$
31	35	28	$n_1 = n_2 = n_3 = 5$
Total $T_1 = 160$	$T_2 = 185$	$T_3 = 480$	$T = 480$
$T_1^2 = 25600$	$T_2^2 = 34225$	$T_3^2 = 18225$	

squares of sum

A	B	C
sum of squares		
5226	6915	3669

Grand Total $= \sum_i \sum_j x_{ij}^2 = 15810$

$$CF = \frac{T^2}{N} = \frac{(480)^2}{15} = 15360$$

$$TSS = \sum_i \sum_j x_{ij}^2 - CF$$

$$= 15810 - 15360 = 450$$

$$SST = \frac{T_1^2}{n1} + \frac{T_2^2}{n2} + \frac{T_3^2}{n3} - CF$$

$$= \frac{25600}{5} + \frac{34225}{5} + \frac{18225}{5} - 15360$$

$SST = 250$

$SEE = TSS - SST$

$= 450 - 250$

$SEE = 200$

ANOVA table

source variation	dif	SS	MSS	F-ratio
B/w treatments	2	250	125	F = 7.49
Error	12	200	16.67	
Total	14			

Fcal = 7.49 Ftab = 3.89

Fcal > Ftab ∴ The hypothesis rejected.

02) plot of land per acre production data

	A	B	C
1	6	5	5
2	7	5	4
3	3	3	3
4	8	7	4

F 39 = 4.26

A	B	C
6	5	5
7	5	4
3	3	3
8	7	4

$T_1 = 24$ $T_2 = 20$ $T_3 = 16$ $T = 3600$ SUM = 158

$T_1^2 = 576$ $T_2^2 = 400$ $T_3^2 = 256$

Squares

	A	B	C
	36	25	25
	49	25	20
	9	9	9
	64	49	16
	158	108	66

Grand Total $= \sum_i \sum_j x_{ij}^2 = 332$

$$CF = \frac{T^2}{N} = \frac{3600}{12} = 300 \qquad N = 12 \quad n_1 = n_2 = n_3 = 4$$

$$TSS = \sum_i \sum_j x_{ij}^2 - CF$$

$$= 332 - 300$$

$$= 32$$

$$SST = \sum_i \frac{T_i^2}{n_i} - CF$$

$$SST = 8$$

$$SSE = TSS - SST$$

$$= 24$$

ANOVA Table

Source variation	d.f	SS	MSS	F-ratio
B/W treatement	2	8	4	$F = 1.50$
Error	9	24	2.66	
Total	11			

$F_{cal} = 1.50$ $F_{tab} = 4.26$

$F_{cal} < F_{tab}$ ∴ The hypothesis is accepted.

03) Set up an analysis of variance table for the following two-way design results: per acre production data of wheat in metric tones; varities of fertilizers varities of seeds $F_{2,6} = 5.14$ & $F_{3,6} = 4.76$

	A	B	C
W	6	5	5
X	7	5	4
Y	3	3	3
Z	8	7	4

	A	B	C	T	T^2
	6	5	5	16	256
	7	5	4	16	256
	3	3	3	9	81
	8	7	4	19	361
P	24	20	16	T = 60	
P^2	576	400	256	N = 12	

squares

A	B	C	
36	25	25	N = 12
49	25	16	
9	9	9	
64	49	16	
SUM 158	108	66	

$\sum_i \sum_j x_{ij} = 332$

$CF = \frac{T^2}{N} = \frac{(60)^2}{12} = 300$

$TSS = \sum_i \sum_j x_{ij}^2 - CF$ $\quad SSR = \frac{\sum_i T_i^2}{n_i} - CF$

$= 332 - 300 = 32$ $\quad SSR = 300^{n_i}$

$SSC = 8$

$SSE = TSS - SSC - SSR$

$SSE = 6$

ANOVA

Source	d.f	S.S	MSS	F ratio
Rows	4-1 = 3	SSR = 18	MSR = 6	Fr = 6
Column	3-1 = 2	SSC = 8	MSC = 4	Fc = 4
Error	3x2 = 6	SSE = 6	MSE = 1	
Total	12-1 = 11			

$Fr = 6 > F(3,6) = 4.76$ and

$Fc = 4 < F(2,6) = 19.33$.

D D M M Y Y Y Y

04) solve using latin squre method (given $F_3 = 4.76$)

D-122	A-121	C-123	B-122
B-124	C-123	A-122	D-125
A-120	B-119	D-120	C-121
C-122	D-123	B-121	A-122

→ Subtract by 120

D=2	A=1	C=3	B=2	$T_1=8$ $T_1^2=64$
B=4	C=3	A=2	D=5	$T_2=14$ $T_2^2=196$
A=0	B=(-1)	D=0	C=1	$T_3=0$ $T_3^2=0$
C=2	D=3	B=1	A=2	$T_4=8$ $T_4^2=64$
$P_1=8$	$P_2=6$	$P_3=6$	$P_4=10$	$T=30$
$P_1^2=64$	$P_2^2=36$	$P_3^2=36$	$P_4^2=100$	$N=16$

Sq ures

D=4	A=1	C=9	B=4
B=16	C=9	A=4	D=25
A=0	B=1	D=0	C=1
C=4	D=9	B=1	A=4
SUM 24	20	14	34

Grand total $\sum_i \sum_j x_{ij}^2 = 92$

$C.F = \frac{T^2}{N} = \frac{(30)^2}{16} = 56.25$

$TSS = \sum_i \sum_j x_{ij}^2 - CF$

$= 92 - 56.25$

$TSS = 35.75$

$SSC = \sum_i \frac{P_i^2}{n} = 35.75$

$SSR = \sum_i \frac{T_i^2}{n} - CF = \frac{T_1^2}{n_1} + \frac{T_2^2}{n_2} + \frac{T_3^2}{n_3} - CF$

$SSR = 24.75$

SUM of treatment

observation

A	121	122	120	122
B	122	124	119	121
C	123	123	121	122
D	122	125	120	123

observation					Q	Q^2
A	1	2	0	2	5	25
B	2	4	-1	1	6	36
C	3	3	1	2	9	81
D	2	5	0	3	10	100

$$SST = \frac{\sum Q_i^2}{n_i} - CF$$

$$SST = \frac{25}{4} + \frac{36}{4} + \frac{81}{4} - \frac{100}{4} - 56.25$$

$SST = 4.25$ $\quad c = 4$

$SSE = TSS - SSR - SSC - SST$ $\quad r = 4$

$SSE = 4$

ANOVA Table

Source of variation	S.S	df	MS	F-ratio
b/w Rows	SSR = 24.75	c-1 = 3	MSC = 8.25	Fc = 30.8
b/w columns	SSC = 2.75	r-1 = 3	MSR = 0.916	Fr = 3.43
b/w treatement	SST = 4.25	(c-1)(r-1) = 9	MST = 0.472	Ft = 1.76
Error	SSE = 4	N-1 = 15	MSE = 0.267	

05) Food 1 8 12 19 8 6 11

Food 2 4 5 4 6 9 7

Food 3 11 8 7 13 7 9

using one-way table using a 0.05 % significance level

→ Food 1 8 12 19 8 6 11 $T_1 = 64$ $T_1^2 = 4096$

Food 2 4 5 4 6 9 7 $T_2 = 35$ $T_2^2 = 1225$

Food 3 11 8 7 3 7 9 $T_3 = 45$ $T_3^2 = 3025$

$N = 18$ $T = 154$

Squares

							Sum of squares
Food 1	64	144	361	64	36	121	790
Food 2	16	25	16	36	81	49	223
Food 3	121	64	49	9	49	81	533

Grand total $\sum_i \sum_j x_{ij}^2 = 1546$

$$CF = \frac{T^2}{N} = \frac{(154)^2}{18} = 1317.55$$

$$TSS = \sum_i \sum_j x_{ij}^2 - CF$$

$$= 1546 - 1317.55$$

$$TSS = 228.45$$

$$SST = \sum_i \frac{T_i^2}{n_i} - CF$$

$$SST = 1317.55 \quad k = 3$$

$$SEE = TSS - SST$$

$$SEE = 155$$

ANOVA Table

Source variation	d.f	SS	MSE	F-ratio
B/w treatments	2	SST = 73.45	MST = 36.725	F = 3.554
Error	15	SEE = 155	MSE = 10.333	
Total	17			

$F_{cal} = 3.554$

$F(k-1, N-k) = F(2,15) = 3.68$

$F_{cal} = 3.554 \quad F_{tab} = 3.68$

$F_{cal} < F_{tab}$ ∴ The hypothesis accepted.

06) Solve using Latin-square design

C	B	A	D
25	23	20	20
A	D	C	B
19	19	21	18
B	A	D	C
19	14	17	20
D	C	B	A
17	20	21	15

D D M M Y Y Y Y

C	B	A	D
25	23	20	20
A	D	C	B
19	19	21	18
B	A	D	C
19	14	17	20
D	C	B	A
17	20	21	15

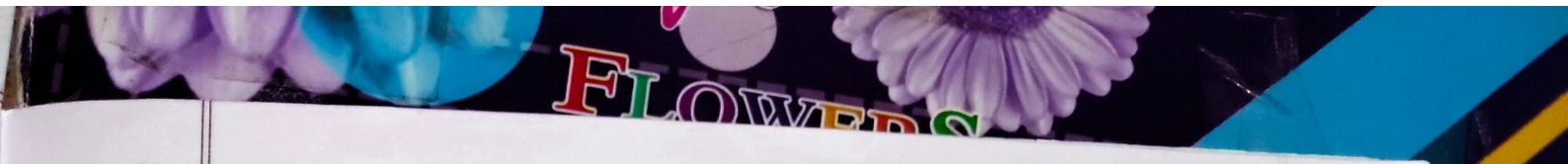
~~-20~~ Total

C	B	A	D	T	T^2
5	3	0	0	8	64
A	D	C	B		
-1	-1	1	-2	-3	9
B	A	D	C		
-1	-6	-3	0	-10	100
D	C	B	A		
-3	0	1	-5	-7	49
P 0	-4	-1	-7	T = -12	
P^2 0	16	1	49	N = 16	

squares

C	B	A	D
25	9	0	0
A	D	C	B
1	1	1	4
B	A	D	C
1	36	9	0
D	C	B	A
9	0	1	25
SUM = 91	46	12	29

$$\Sigma_i \Sigma_j x_{ij}^2 = 122$$



D D M M Y Y Y Y

$CF = \frac{T^2}{N} = \frac{(112)^2}{16} = \frac{1144}{16} = 9$

$TSS = \sum_i \sum_j x_{ij}^2 - CF$

$= 122 - 9$

$= 113$

$SSR = \sum_i \frac{T_i^2}{n_i} - CF$

$= 55.5 - 9$

$= 46.5$

$SSC = 0 + \frac{16}{4} + \frac{1}{4} + \frac{49}{4} - 9$

$SSC = 7.5$

Sum of treatments					Q	Q^2
A	0	-1	-6	-5	-12	144
B	3	-2	-1	1	1	1
C	5	1	0	0	6	36
D	0	-1	-3	-3	-7	49

$SST = \sum \frac{Q_i^2}{n_i} - CF$

$SST = 48.5$

$SSE = TSS - SST - SSC$

$SSE = 10.5$

ANOVA Table

Source	d.f	SS	MSS	F ratio
row	3	SSR = 46.5	MSR = ~~15.5~~ 2.5	Fr = ~~0.771~~ 1.857
column	3	SSC = 7.5	MSC = ~~1.664~~ 16.67	Fc = ~~2.37~~ 1.42
treatment	3	SST = 48.5	MST = ~~3.064~~	FT = 9.23
Error	~~6~~ 6	SSE = 10.5	MSE = ~~1.75~~	
	~~8~~			

DD MM YYYY

07) Set up ANOVA for the following information relating to three drugs testing to judge the effectiveness in reducing blood pressure for three different groups of people

Group of people	Drug X	Y	Z
A	14	10	11
	15	9	11
B	12	7	10
	11	8	11
C	10	11	8
	11	11	7

Do the drugs act differently?
Are the different groups of people affected differently?
Is the interaction term significant?
Answer the above questions taking a significant level of 5%.

Group of people	Drug X	Y	Z	sum (T)	T^2
A	14	10	11	$T_1 = 70$	$T_1^2 = 4900$
	15	9	11		
B	12	7	10	$T_2 = 59$	$T_2^2 = 3481$
	11	8	11		
C	10	11	8	$T_3 = 58$	$T_3^2 = 3364$
	11	11	7		
Sum P	73	56	58	= 187	—
squares P^2	5329	3136	3364	—	—

sum squares

Group of people	Drug X	Y	Z	sum of squares
A	196	100	121	844
	225	81	121	

144	49	100	599
121	64	121	
100	121	64	576
121	121	49	

Grand Totals $\sum_i \sum_j x_{ij}^2 = 2019$

$H_0 = \mu_1 = \mu_2 = \mu_3$

$C.F = \frac{T^2}{N} = \frac{(187)^2}{18} = 1942.722$

$TSS = \sum_i \sum_j x_{ij}^2 - CF$

$= 2019 - 1942.722$

$TSS = 76.28$

$SSR = \sum_i \frac{T_i^2}{n_i} - CF$

$= \frac{4900}{6} + \frac{3481}{6} + \frac{33364}{6} - 1942.722$

$= 14.78$

$SSC = \sum_i \frac{P_i^2}{n_i} - CF$

$= 888.16 + 522.66 + 560.67 - 1942.722$

$SSC = 28.77$

Mean $= \frac{14+15}{2} = 14.5, \frac{10+9}{2} = 9.5, \frac{11+11}{2} = 11$

$\frac{11+12}{2} = 11.5, \frac{7+8}{2} = 7.5, \frac{10+5}{2} = 10.5$

Sum of squares within Treatments (SSW)

$(14-14.5)^2 + (15-14.5)^2 + (10-9.5)^2 + (9-9.5)^2 + (11-11)^2 + (11-11)^2 + (11-11)^2 + (12-11.5)^2 + (11-11.5)^2 + (7-7.5)^2 + (8-7.5)^2 + (10-10.5)^2 + (11-10.5)^2 + (10-10.5)^2 + (11-10.5)^2 + (11-11)^2 + (11-11)^2 + (8-7.5)^2 + (7-7.5)^2$

$SSW = 3.50$

D D M M Y Y Y Y

$SSI = TSS - SSR - SSC - SSW$

$= 76.28 - 14.78 - 28.77 - 3.50$

$= 29.23$

ANOVA

Sources of variation	d.f	SS	MSS	F-ratio	Conclusion
Rows	$r-1$	$SSR =$	$MSR = \frac{14.78}{2}$	$F_r = 19$	$F(2,9) = 4.26$
	$3-1=2$	14.78	$= 7.39$		$F_r > F(2,9)$
					H_0 rejected
Columns	$c-1$	$SSC =$	$MSC = 14.385$	$F_C = 3.7$	$F_C > F_{(2,9)}$
	$3-1=2$	28.77			H_0 rejected
Due to Interaction	$(r-1)(c-1)$	$SSI =$	$MSI = 7.33$	$F_I = 18.84$	$F_I > F(4,9)$
	4	29.23			H_0 rejected
within treatment error	(N - total mean)	$SSW =$	$MSW = 0.389$	-	-
	$18-9=9$	3.50			
Total	$N-1=17$				

∴ The drugs act differently.

∴ The different groups the people are affected differently

* Interaction term is significant.

Because from the above table, all the three F-ratios are significant at 5% level of significance.

(8) Construct one-way Table for

Fertilizer -1	6	8	4	5	3	4
Fertilizer -2	8	12	9	11	6	8
Fertilizer -3	13	9	11	8	7	12

-> for constructing one-way

							Total	squares
Fertilizer -1	6	8	4	5	3	4	$T_1 = 30$	$T_1^2 = 900$
Fertilizer -2	8	12	9	11	6	8	$T_2 = 54$	$T_2^2 = 2916$
Fertilizer -3	13	9	11	8	7	12	$T_3 = 60$	$T_3^2 = 3600$
							$N = 18$	$T = 144$

squares

							Sum
Fertilizer 1	36	64	16	25	9	16	166
2	64	144	81	121	36	64	510
3	169	81	121	64	49	144	628

$$\sum_i \sum_j x_{ij}^2 = 1304$$

$H_0 = \mu_1 = \mu_2 = \mu_3$

$$CF = \frac{T^2}{N} = \frac{144}{18} = 1152$$

$$TSS = \sum_i \sum_j x_{ij}^2 - CF$$

$$= 1304 - 1152$$

$$= 152$$

$$SST = \left(\frac{T_1^2}{n_1} + \frac{T_2^2}{n_2} + \frac{T_3^2}{n_3}\right) - CF$$

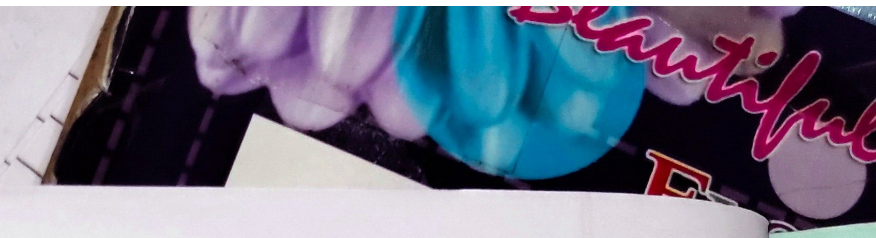
$$= \left(\frac{900 + 2916 + 3600}{6}\right) - 1152$$

$SST = 84$

$SEE = TSS - SST$

$= 152 - 84$

$SEE = 68$ $\quad k = 3$



D D M M Y Y Y Y

ANOVA

Source variation	d.f	SS	MSS	f-ratio
B/w treatments	2	SST = 84	MST = 42	F = 9.265
Error	15	SEE = 68	MSE = 4.53	
Total	17			

$F(K-1, N-K) = F(2,15) = 3.68$

$F_{cal} > F(2,15)$

$\therefore$ The hypothesis rejected.